

# Solve problems involving money and measures

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains? Copy or complete the first step.

2. A 12 km route is split into 8 equal stages. How many metres per stage?

3. A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?

4. Miro says: Miro mixes pounds and pence or kilometres and metres in one calculation. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains? Copy or complete the first step.  
**Answer** £1,132.
2. A 12 km route is split into 8 equal stages. How many metres per stage?  
**Answer** 1,500 m.
3. A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?  
**Answer** 1,000 litres.
4. Miro says: Miro mixes pounds and pence or kilometres and metres in one calculation. Is this correct? Explain.  
**Answer** Convert to one unit before calculating and label the final unit.

# Solve problems involving money and measures

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?

6. Check this result using an inverse, estimate or known fact: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?

2. A 12 km route is split into 8 equal stages. How many metres per stage?

7. A £6,000 budget must cover 48 identical items and leave at least £720. What is the greatest whole-pound price per item?

3. A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?

8. Identify and correct this misconception: Miro mixes pounds and pence or kilometres and metres in one calculation.

4. Solve this independently and show every stage: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A 12 km route is split into 8 equal stages. How many metres per stage?

10. Write and solve a similar question to: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?

# Teacher answers and checking prompts

1. A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?  
**Answer** £1,132.
2. A 12 km route is split into 8 equal stages. How many metres per stage?  
**Answer** 1,500 m.
3. A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?  
**Answer** 1,000 litres.
4. Solve this independently and show every stage: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?  
**Answer** £1,132.
5. Use a second method or representation for: A 12 km route is split into 8 equal stages. How many metres per stage?  
**Answer** Expected result: 1,500 m. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?  
**Answer** Expected result: 1,000 litres. A valid check must be shown.
7. A £6,000 budget must cover 48 identical items and leave at least £720. What is the greatest whole-pound price per item?  
**Answer** £110.
8. Identify and correct this misconception: Miro mixes pounds and pence or kilometres and metres in one calculation.  
**Answer** Convert to one unit before calculating and label the final unit.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Solve problems involving money and measures

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. A £6,000 budget must cover 48 identical items and leave at least £720. What is the greatest whole-pound price per item?

6. Create a new example that tests the same idea as: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?

2. Prove the answer to this question, rather than only calculating it: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A 12 km route is split into 8 equal stages. How many metres per stage?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 1,000 litres.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro mixes pounds and pence or kilometres and metres in one calculation.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. A £6,000 budget must cover 48 identical items and leave at least £720. What is the greatest whole-pound price per item?  
**Answer** £110.
2. Prove the answer to this question, rather than only calculating it: A class has a £2,500 budget and buys 18 tablets cases at £76 each. How much remains?  
**Answer** Expected result: £1,132. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A 12 km route is split into 8 equal stages. How many metres per stage?  
**Answer** Expected result: 1,500 m. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 1,000 litres.  
**Answer** One valid reconstruction is: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro mixes pounds and pence or kilometres and metres in one calculation.  
**Answer** Convert to one unit before calculating and label the final unit.
6. Create a new example that tests the same idea as: A tank contains 4,500 litres. It fills 28 containers of 125 litres. How much remains?  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.